

Feathers as a source of RNA for genomic studies in avian species

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Abstract Dioxins and dioxin-like chemicals (DLCs) cause a suite of adverse effects in terrestrial species. Most of the adverse effects occur subsequent to binding to the aryl hydrocarbon receptor. Avian species vary in their sensitivity to the effects of DLCs and current research indicates that this is mediated by variations in the amino acid sequence within the ligand binding domain (LBD) of the aryl hydrocarbon receptor 1 (AHR1). Eighty-eight avian species have been classified into three broad categories of sensitivity, based on the amino acid variations within the AHR1 LBD: sensitive type 1 (Ile324_Ser380), moderately sensitive type 2 (Ile324_Ala380), and relatively insensitive type 3 (Val324_Ala380). Risk assessment of avian species can be complicated due to the variability in sensitivity among species. A predictive tool for selecting the priority species at a given site would have broad implications for the risk assessment community. We present a method for AHR1 genotyping using plucked feathers as a source of RNA. The method is extremely robust, requires minimal sample processing and handling, and eliminates the need for blood sampling or tissue collection from the species of interest. Using this method we were able to determine the amino acid sequence of the AHR LBD of three avian species: the chicken, the herring gull, and the zebra finch, and to categorize them based on the identity of amino acids at key sites within the LBD.

Keywords Aryl hydrocarbon receptor · RNA · Nucleic acid extraction · Feather · Bird · Sequencing · Dioxin

Introduction

Dioxin and dioxin-like compounds (DLCs) are ubiquitous in the environment and cause a suite of well characterized adverse effects in terrestrial species, including but not limited to, tumor promotion, teratogenicity, wasting, alterations in endocrine homeostasis, and induction or repression of a large number of genes (Blankenship et al. 2003; Cantrell et al. 1996; Kopf and Walker 2009; Nebert et al. 2004; Peterson et al. 1993; Vezina et al. 2004). Avian species vary dramatically in their sensitivity to the effects of DLCs (Hoffman et al. 1998; Brunstrom 1986; Kennedy et al. 1994), and as many of the toxic and biochemical effects are elicited subsequent to binding to the aryl hydrocarbon receptor (AHR) (Carney et al. 2006; Denison et al. 2011; Hahn 1998), considerable research of late has focussed on investigating its role as a moderator of the toxic and biochemical effects (Farmahin et al. 2012; Hahn et al. 2006; Karchner et al. 2000; Manning et al. 2012; Mimura and Fujii-Kuriyama 2003; Okey et al. 2005). Avian species possess two isoforms of AHR, AHR1 and AHR2, (Yasui et al. 2007) and research has revealed that knowledge of the amino acid sequence of the ligand binding domain (LBD) of AHR1 can be useful in predicting the sensitivity of birds to the effects of DLCs (Farmahin et al. 2013; Head et al. 2008; Manning et al. 2012). The LBD has six variable amino acids and variations at positions 324 and 380, can be used to classify birds into three main types: sensitive type 1 (Ile324_Ser380); moderately sensitive type 2 (Ile324_Ala380); relatively insensitive type 3 (Val324_Ala380) (Cohen-Barnhouse et al. 2011; Farmahin et al. 2013; Head et al. 2008; Hervé et al. 2010; Manning et al. 2012). Using this methodology, 88 species have been classified with some representative examples being the chicken and European starling classified as type 1, the ring-necked pheasant, blue jay and American crow classified as

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type 2 and the herring gull, bald eagle and mallard duck classified as type 3 (see Farmahin et al. 2013 for a full list).

One of the goals of this laboratory is to develop methods that will improve our understanding of avian species sensitivity to the effects of DLCs to aid in risk assessment. Current protocols for sequencing the avian AHR1 often require euthanasia of the animal of interest, and although non-lethal methods for sequencing from blood have been developed (Head et al. 2010), the rate of success is limited (Jones et al., un-published data from our laboratory during the past 3 years). Non-lethal sampling methods are beneficial not only from an ethical perspective but also, when working with endangered or at risk species, permits may be prohibitive to lethal sampling methods. Furthermore, a high rate of success is clearly desirable. Feathers are frequently used for DNA genotyping and molecular sexing, as well as for contaminant and stable isotope analysis (Bush et al. 2005; Harvey et al. 2006; Hobson et al. 2004, 2012; Hobson and Wassenaar 1997; Horvath et al. 2005; Jaspers et al. 2007; Jensen et al. 2003; Ramos et al. 2009; Segelbacher 2002; Taberlet and Bouvet 1991). More recently, dried feather shafts have been used to detect avian leukosis virus and avian bornavirus by RT-PCR (de Kloet et al. 2011; Hatai et al. 2009).

In this study we assessed the utility of freshly plucked feathers as a source of RNA for sequencing of the AHR1 LBD of three avian species. We tested three main parameters: sample storage, RNA yield and sample stability. The results indicate that feathers are a useful sample source for AHR sequencing as they provide high RNA yield, the RNA is stable at room temperature and feathers can be shipped long distances without the need for specialized handling. This improves the application of AHR sequencing for larger scale studies where facilities for storage and sample handling may be limited, and it will enable determination of the AHR1 genotype of a large number of species from varied locales, with minimal stress to the animal of interest. The chicken (*Gallus gallus*) and the herring gull (*Larus argentatus*) were selected for method validation because feathers from these species were easy to acquire and their AHR LBD sequences are well documented. The zebra finch (*Taeniopygia guttata*) was selected because it is a commonly used test species, feathers were readily available and the AHR1 LBD sequence was unknown.

Materials and methods

Sample preparation

Secondary covert feathers were plucked from juvenile herring gulls at various sites on the Great Lakes in Ontario, Canada and were placed immediately in liquid nitrogen (LN₂), or in paper envelopes at ambient temperature.

Samples collected in LN₂ were transferred to a −80 °C freezer upon receipt at the laboratory, while samples collected at ambient temperature were placed in a 4 °C refrigerator upon receipt at the laboratory. Some samples from LN₂ were transferred to RNeasy (Invitrogen, Burlington, ON, Canada) and stored at 4 °C for several days.

Secondary covert feathers were collected from juvenile chickens at the Canadian Food Inspection Agency (Ottawa, ON, Canada). Feathers were placed in paper envelopes and transported at ambient temperature. Samples were transferred to a 4 °C refrigerator upon receipt at the laboratory.

Tail feathers were collected from juvenile zebra finches at Simon Fraser University (Vancouver, BC, Canada), placed in paper envelopes and shipped at ambient temperature. Samples were transferred to a 4 °C refrigerator upon receipt at the laboratory.

RNA isolation and reverse transcription

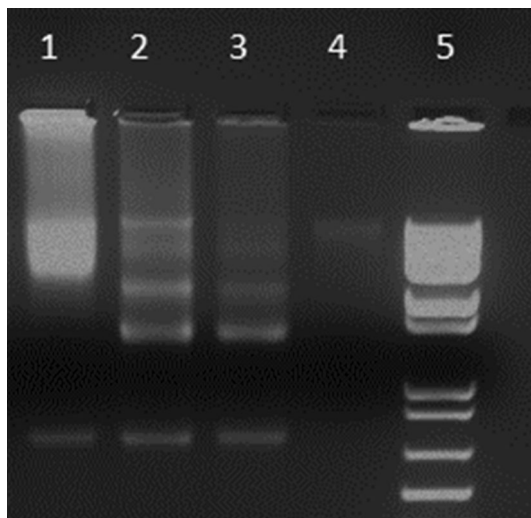
A 0.5–1 cm piece of the feather calamus including the blood clot was cut away from the feather. When the feather shaft was soft enough, the pulp inside was squeezed out and used for RNA isolation. If the shaft was too hard to cut or if cutting it would have resulted in a piece smaller than 0.5 cm, the entire piece was used. All samples were put into 350 µl of buffer RLT (Qiagen, Mississauga, ON, Canada) in a 2 ml tube containing a stainless steel ball. A Retsch Mixer Mill 200 (Verden Scientific Inc., Newtown, PA, USA) was used to homogenize the sample for up to 4 min at 20 Hz. After homogenization, 1 volume of 70 % ethanol was added, the sample was mixed by pipetting and transferred to an RNeasy spin column (Qiagen). RNA was extracted according to the manufacturer's protocol (RNeasy Minikit, Qiagen), with a final elution in 30–50 µl of RNase-free water. RNA concentration was measured using a NanoDrop 2000 (ThermoFisher Scientific, Ottawa, ON, Canada). Up to 2 µg of RNA was reverse transcribed into cDNA using Superscript II (Invitrogen, Burlington, ON, Canada) and random hexamers (Invitrogen) according to the manufacturer's instructions. In order to prevent cross-contamination, tools and workspace were rigorously treated with DNA-away (Qiagen) and UV irradiation, and individual species were processed on different days.

Polymerase chain reaction (PCR)

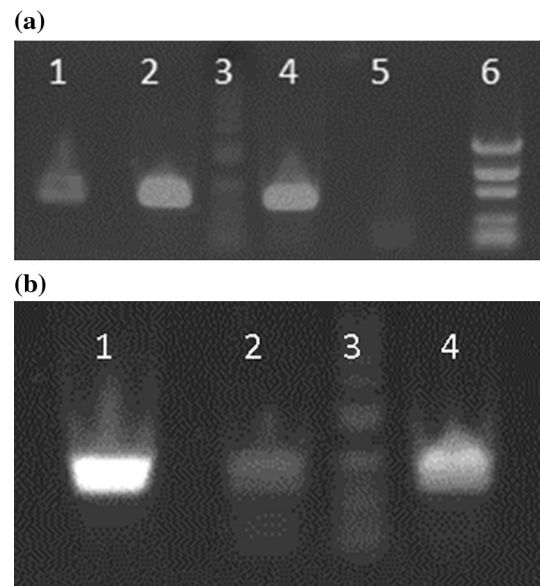
Following reverse transcription, the KAPA HiFi HotStart kit (KAPA Biosystems, Boston, MA, USA) was used in 50 µl PCR reactions with 2 µl of cDNA in an MJ Research PTC 200 thermal cycler. The primers were designed based on the sequence of the chicken AHR1 LBD, and have been used to identify approximately 90 avian species to date.

Table 1 Summary of storage conditions and duration of storage on RNA yield from feather samples

Species	Sample number	Collection date	RNA extraction date	Storage conditions	RNA yield (μg)	PCR success (y/n)
Chicken	1	28/01/2014	10/02/2014	Dry, 4 °C	6.3	y
	2		10/02/2014	Dry, 4 °C	9.8	y
	3		10/02/2014	Dry, 4 °C	13.4	y
	4		02/04/2014	Dry, 4 °C	8.0	y
	5		02/04/2014	Dry, 4 °C	9.8	y
Herring gull	1	13/06/2013	18/11/2013	Dry, 4 °C	4.4	y
	2		20/11/2013	RNA later, 4 °C	13.7	y
	3		20/11/2013	LN ₂ , -80 °C	207	y
	4		25/11/2013	RNA later, 4 °C	21.1	y
	5		09/04/2014	Dry 4 °C	1.9	y
Zebra finch	1	02/20/2014	02/26/2014	Dry, 4 °C	1.5	y
	2		02/28/2014	Dry, 4 °C	0.5	y
	3		02/28/2014	Dry, 4 °C	2.6	y
	4		02/28/2014	Dry, 4 °C	0.3	y
	5		02/28/2014	Dry, 4 °C	0.4	y
	6		09/04/2014	Dry, 4 °C	0.4	y

**Fig. 1** Effect of storage conditions on RNA quality. RNA samples were run on a 1.5 % agarose gel with ethidium bromide staining. Lane 1: chicken feather stored dry at 4 °C (2.7 μg RNA); lane 2: herring gull feather stored in LN₂ (2 μg RNA); lane 3: herring gull feather stored in RNAlater (2.7 μg RNA); lane 4: zebra finch feather stored dry at 4 °C (418 ng RNA); lane 5: 1 Kb ladder

(Farmahin et al. 2013). The forward primer was: CCAGA CCAACTTCCTCCAGA. The Reverse primer was: ATGT TTGCCACTGGTG. The PCR program used was 2 min at 95 °C, 35 cycles of: 20 s at 98 °C, 30 s at 62 °C, 1 min at 72 °C, followed by 2 min at 72 °C, with a final hold at 4 °C. Products were analyzed on 0.8 % E-Gels (Invitrogen) and collected in 20 μl of water. Samples were quantified against a quantitative DNA mass ladder (Invitrogen). Sequencing was done with an ABI 3130 XL DNA

**Fig. 2** Effect of sample storage conditions on AHR1 LBD PCR products from feathers (0.8 % E-Gel). **a** Herring gull feathers lane 1: 4 months dry at 4 °C; lane 2: 4 months at -80 °C; lane 3 marker; lane 4: RNAlater; lane 5: NTC; lane 6: Low DNA mass ladder. **b** lane 1: chicken feather stored dry at 4 °C for 5 weeks; lane 2: herring gull feather stored dry at 4 °C for 10 months; lane 3: 1 Kb ladder; lane 4: zebra finch feather stored dry at 4 °C for 6 weeks

Analyzer (LifeTechnologies, Burlington, ON, Canada) at the Institute de Biologie Intégrative et de Systèmes (University of Laval, Laval, QC). Sequencer software (Gene Codes Corp., Ann Arbor, MI, USA ver. 4.9) was used to align DNA sequences. A consensus sequence for each species was obtained by aligning all sample sequences.

Fig. 3 Zebra finch multiple sequence alignment demonstrating 100 % agreement between four individuals

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1 CTGTGTATGAGAGGAAGTGGGTACCAGTTTATTCATGCAGCTGACATGCTTTATTGTGCT
2 CTGTGTATGAGAGGAAGTGGGTACCAGTTTATTCATGCAGCTGACATGCTTTATTGTGCT
3 CTGTGTATGAGAGGAAGTGGGTACCAGTTTATTCATGCAGCTGACATGCTTTATTGTGCT
4 CTGTGTATGAGAGGAAGTGGGTACCAGTTTATTCATGCAGCTGACATGCTTTATTGTGCT
*****
1 GAAAATCACATCCGAATGATGAAGACAGGTGAGAGTGAATGACTGTATTTAGGCTTCTA
2 GAAAATCACATCCGAATGATGAAGACAGGTGAGAGTGAATGACTGTATTTAGGCTTCTA
3 GAAAATCACATCCGAATGATGAAGACAGGTGAGAGTGAATGACTGTATTTAGGCTTCTA
4 GAAAATCACATCCGAATGATGAAGACAGGTGAGAGTGAATGACTGTATTTAGGCTTCTA
*****
1 ACCAAAGAAAATCGATGGGCCTGGGTACAGGCAAATGCACGGCTTGTCTACAAAAATGGA
2 ACCAAAGAAAATCGATGGGCCTGGGTACAGGCAAATGCACGGCTTGTCTACAAAAATGGA
3 ACCAAAGAAAATCGATGGGCCTGGGTACAGGCAAATGCACGGCTTGTCTACAAAAATGGA
4 ACCAAAGAAAATCGATGGGCCTGGGTACAGGCAAATGCACGGCTTGTCTACAAAAATGGA
*****
1 AGACCAGATTACATCATTGCCACGCAAAGACCTCTTACAGATGAAGAAGGGGCAGAACAT
2 AGACCAGATTACATCATTGCCACGCAAAGACCTCTTACAGATGAAGAAGGGGCAGAACAT
3 AGACCAGATTACATCATTGCCACGCAAAGACCTCTTACAGATGAAGAAGGGGCAGAACAT
4 AGACCAGATTACATCATTGCCACGCAAAGACCTCTTACAGATGAAGAAGGGGCAGAACAT
*****
1 CTACGGAAGCGTAACATGAAGTTGCCCTTC
2 CTACGGAAGCGTAACATGAAGTTGCCCTTC
3 CTACGGAAGCGTAACATGAAGTTGCCCTTC
4 CTACGGAAGCGTAACATGAAGTTGCCCTTC
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Fig. 4 Multiple sequence alignment of aryl hydrocarbon receptor amino acid sequences determined from feathers of chicken, herring gull and zebra finch. **Bold letters** and **colon** indicate differences between species (ZEFI zebra finch; HERG herring gull)

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3 HERG  AMNFQRLKFLHGQKKKGKDATLSQLALFAVATPLQPPSILEIRTKNFIRTKHKLDF
2 ZEFI  AMNFQRLKFLHGQKKKGKDATLSQLALFAVATPLQPPSILEIRTKNFIRTKHKLDF
1 Chicken AMNFQRLKFLHGQKKKGDGALSQLALFAVATPLQPPSILEIRTKNFIRTKHKLDF
*****
3 HERG  TPTGCDAKGKIVLGYTEAELCMRGTGYQFVHAADMLYCAENHVRMMKTGESGMTVFRLLT
2 ZEFI  TPTGCDAKGKIVLGYTEAELCMRGTGYQFIHAADMLYCAENHIRMKTGESGMTVFRLLT
1 Chicken TPTGCDAKGKIVLGYTEAELCMRGTGYQFIHAADMLYCAENHVRMMKTGESGMTVFRLLT
*****
3 HERG  KENRWAWQANARLVYKNGRPDYIIATQRPLTDEEGAHLRKRNMKLP
2 ZEFI  KENRWAWQANARLVYKNGRPDYIIATQRPLTDEEGAHLRKRNMKLP
1 Chicken KENRWAWQANARLVYKNGRPDYIISTQRPLTDEEGAHLRKRNMKLP
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Results

RNA extraction

Effects of sample storage methods on RNA yield and stability are summarized in Table 1. RNA quality was assessed by agarose gel electrophoresis and visualization with ethidium bromide staining (Fig. 1). Herring gull feathers preserved immediately in LN₂ and then transferred to a −80 °C freezer yielded 200 µg of RNA after approximately 4 months of storage at 4 °C. Samples refrigerated in RNA-later yielded 14–21 µg, while the yield from samples stored dry at 4 °C for 4 months was 4 µg. Feathers stored dry for up to 10 months at 4 °C exhibited a 100 fold decrease in RNA yield (1.9 µg) compared to samples stored at −80 °C.

Chicken feathers placed in paper envelopes and stored dry in a refrigerator for up to 2.5 months yielded 6–13 µg of RNA. Zebra finch feathers placed in paper envelopes and stored dry in a refrigerator for up to 6 weeks yielded from 0.3 to 2.6 µg of RNA. Neither chicken nor zebra finch feathers exhibited a decrease in RNA yield over the time period assessed.

PCR and sequencing

All samples produced a single PCR product, which was quantified either against a DNA mass ladder or by use of the NanoDrop 2000 (Fig. 2a, b). Storage of herring gull samples for up to 10 months at 4 °C reduced the amount of PCR product (Fig. 2b), but had no effect on sequencing. Multiple individuals from each species were analyzed and

sequences were aligned (Fig. 3). Amino acid variations between species were readily determined (Fig. 4) and each species was classified based on its respective amino acid sequence at positions 324 and 380: chicken type 1 (Ile324_Ser380); zebra finch type 2 (Ile324_Ala380); herring gull type 3 (Val324_Ala380). The zebra finch sequence was submitted to GenBank (accession number KJ825952).

Discussion

Avian species can exhibit up to 1000-fold differences in their sensitivity (Head et al. 2008) to the toxic and biochemical effects of DLCs, therefore when considering ecological risk assessments, it may be difficult to determine which species may be most at risk at any given site. A key requirement is to select species that will be representative of the range of sensitivities, in order that the assessment will be neither over-nor under-protective to all species present. As such, knowledge of the amino acid sequence of avian AHR1 has potential as a predictive tool to aid in priority species selection, as sampling every organism at every site is unrealistic both from a time and money management perspective. This strategy was recently used in several ecological risk assessments of the Tittabawasee River system in Midland, Michigan, USA (Fredricks 2009; Seston 2010). Coupled with knowledge of contaminant levels, sequencing of the avian AHR1 LBD may aid a wide scientific community in directing future risk assessment strategies.

Feathers have proven to be a useful non-lethal and minimally invasive sample source for a variety of scientific methods (Bush et al. 2005; Harvey et al. 2006; Hobson and Wassenaar 1997; Jaspers et al. 2007). Feathers plucked during an active growth or regeneration state contain both blood and residual epithelial cells in the pulp therefore they are useful as a source of RNA for genetic analysis. To our knowledge, this is the first report of feathers being used as a sample source for RNA genotyping, and this could allow for characterization of the expression of other genes in the dioxin response pathway, or other biochemical pathways. The RNA may also be used to detect a specific disease state (de Kloet et al. 2011; Hatai et al. 2009) or to detect genes that are representative of phenotypic variation within populations (Li et al. 2012).

In general, dried feathers required no specialized storage and exhibited minimal decrease in RNA yield after greater than two months when stored at 4 °C. This provides a dramatic improvement to current methods of using blood either on filter paper or preserved in RNAlater (Head et al. 2010), and tissues flash frozen in liquid nitrogen immediately after dissection. The high yield of RNA from feather samples provides ample material for any subsequent

genetic analysis, which decreases the handling and sampling of individual animals. In contrast to Head et al. (2010), we were able to characterize the AHR1 LBD directly from PCR products without the need for cloning, due in part to recent advances in molecular biology techniques, which also decreased sample processing time to less than 8 h. The use of RNA is preferred over DNA for similar reasons as it requires only one round of PCR versus up to four for DNA, thus further decreasing sample handling and processing time. The method presented here is extremely robust and minimally labour intensive and could prove to be a valuable tool for field researchers who may be working in difficult to access locations with minimal facilities for storage of samples.

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Conflict of interest The authors declare that they have no conflict of interest.

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